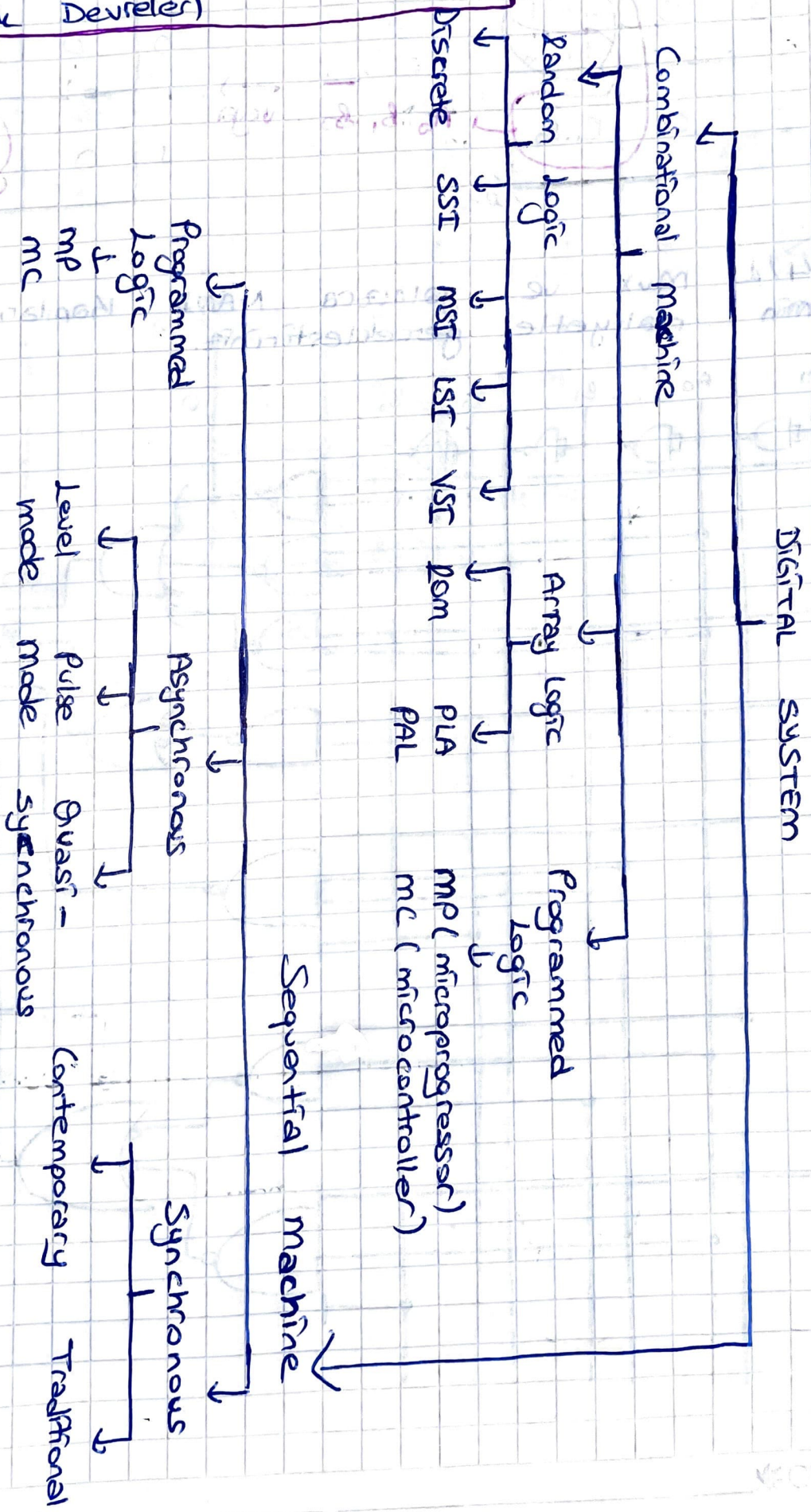
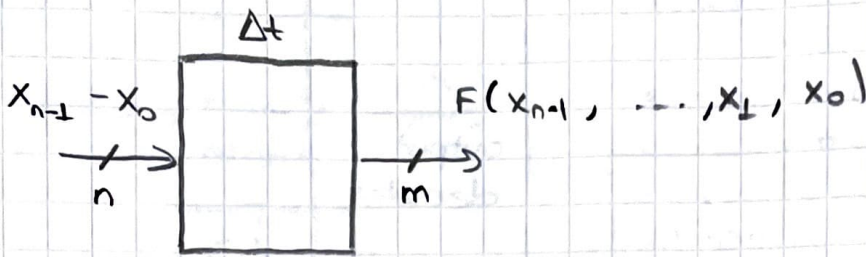


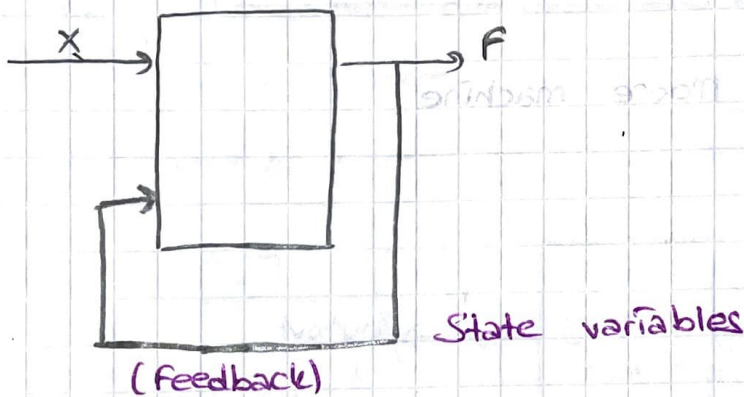
SEQUENTIAL LOGIC CIRCUITS (MACHINES) (Ardisil Logik Devreler)



Combinational Logic Devices Properties



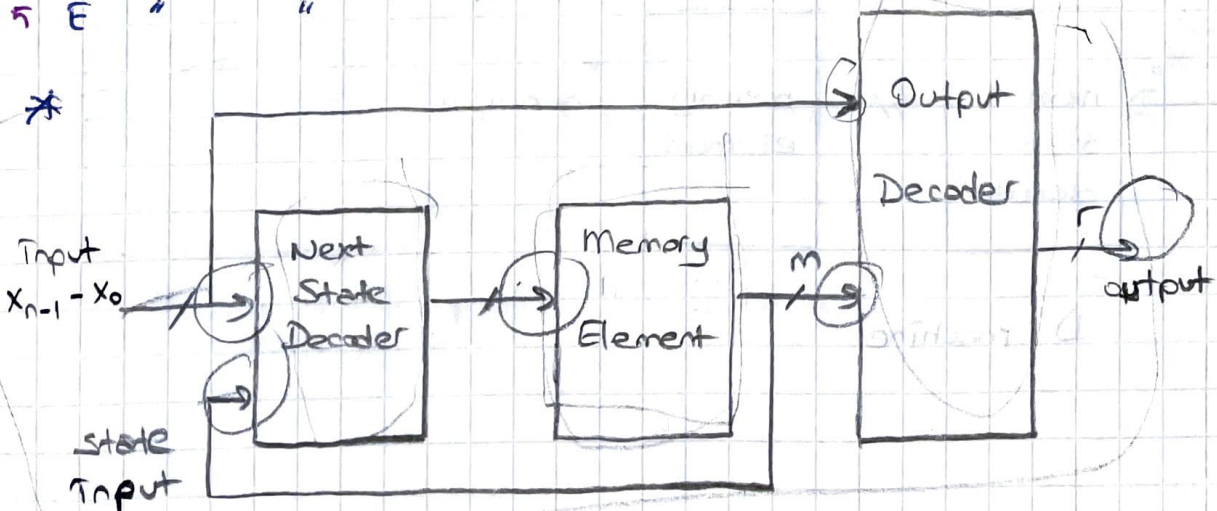
Sequential Devices Properties



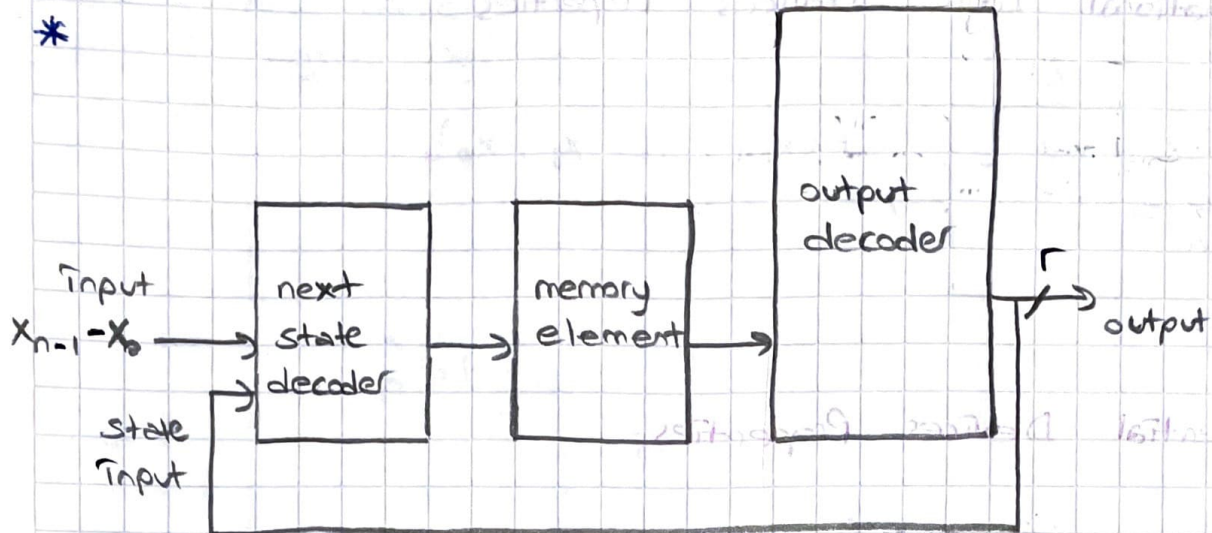
Models for Sequential Machines

- 1) A Type Machine = mealy machine
- 2) B " " = moose "
- 3) C " " = " "
- 4) D " " = Look-up "
- 5 E " "

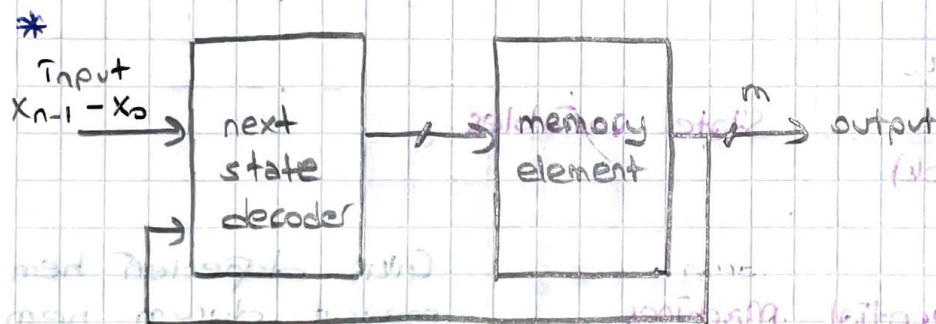
Giris degerleri hem mevcut durum hem de giris degerleri taraf. belirlenir.



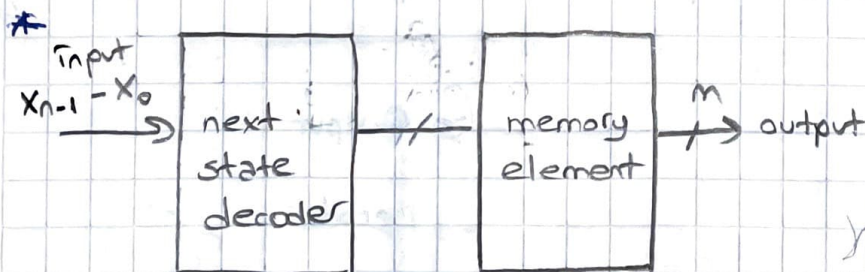
A machine : Mealy machine



B machine: Moore machine

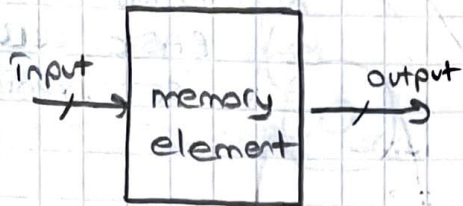


C machine: Moore machine



D machine

*



E machine

→ input bir sonraki durumu etkiler

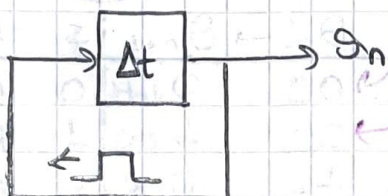
→ mevcut çıkış değerleri yalnızca mevcut durum tarafı belirlenen bir sonlu durum makinesidir.

BASIC MEMORY CELL

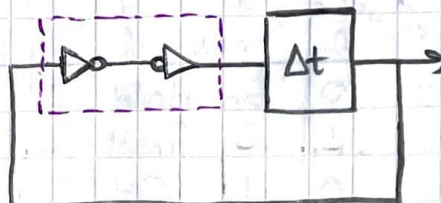
→ Heuristic development of the Basic cell

NSD (Next State Decoder)

Delay Element



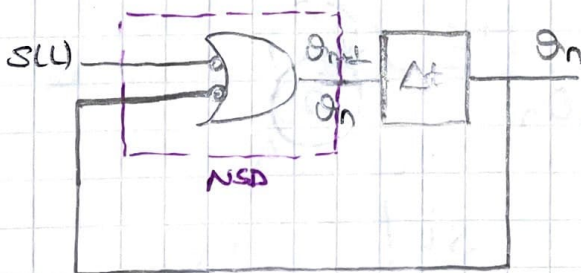
$$Q_{n+1} = Q_n = 1$$



$$Q_{n+1} = Q_n = 1$$

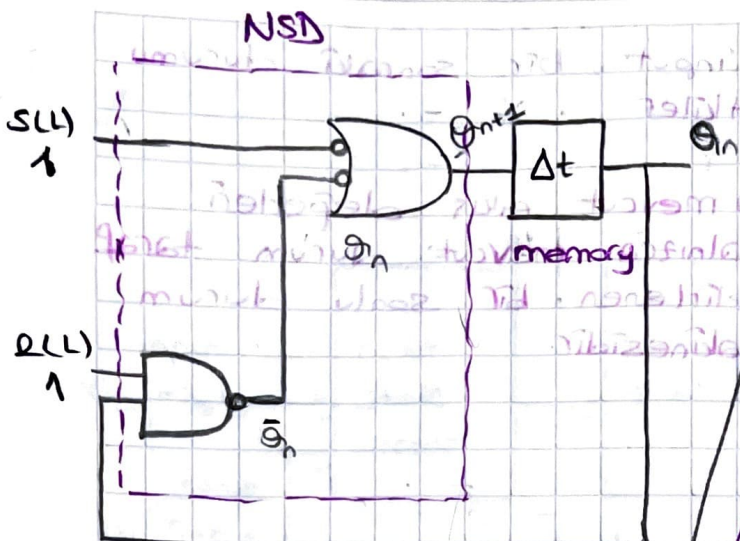
Ideal connecting wires

A) NAND-CENTERED BASIC CELL



$$Q_{n+1} = S * Q_n$$

S	Q_{n+1}	
0	Q_n	Hold
1	1	Set

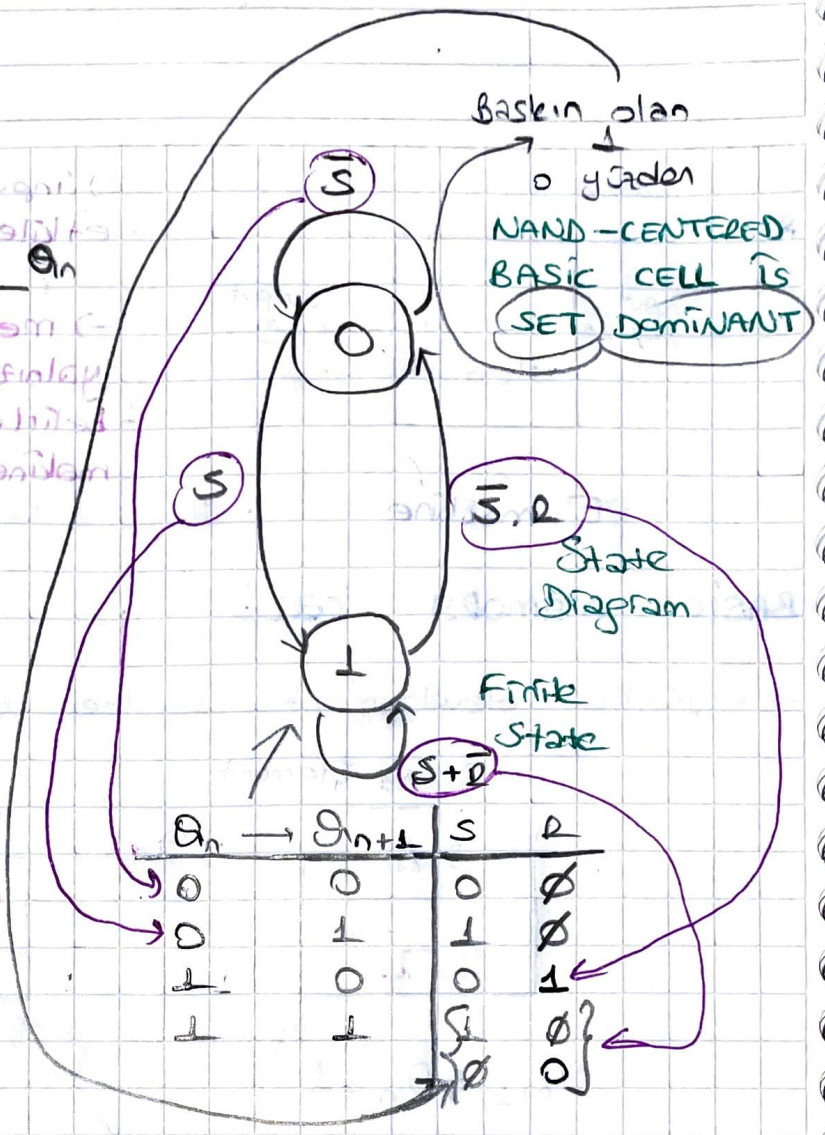


$Q_{n+1} = S + \bar{R} \cdot Q_n$

setleme ucu resetleme ucu

S	R	Q_{n+1}	
0	0	Q_n	Hold
0	1	0	Reset
1	0	1	Set
1	1	1	Set *

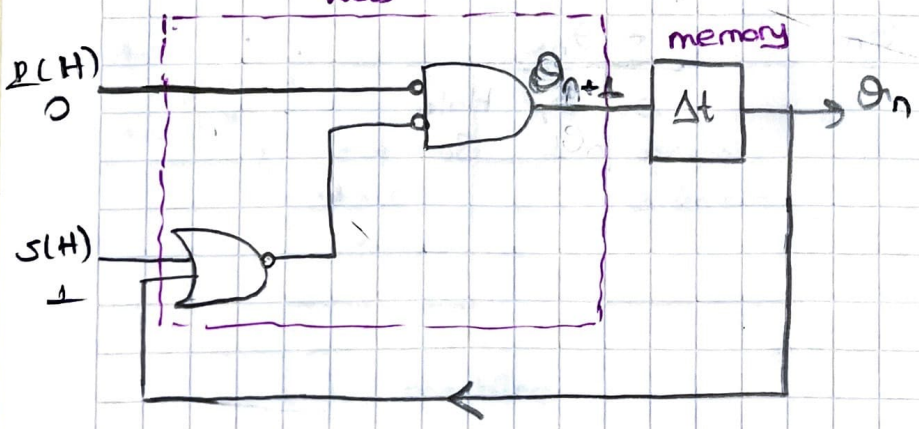
Operation Table
(Truth Table)



$Q_n \rightarrow Q_{n+1}$	S	R
0 → 0	0	0
0 → 1	1	0
1 → 0	0	1
1 → 1	1	0

State Transition Table
(Excitation Table)

B) NOR-CENTERED BASIC CELL

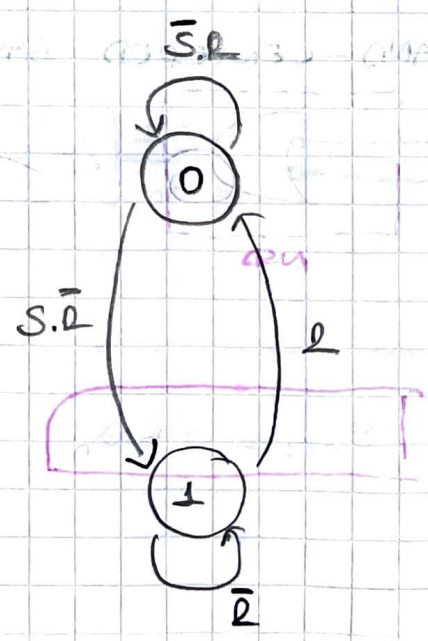


Giriler 1'de aktif

S	R
0	1

olur.
yani girislerin zıtını aldık

$\frac{R(H)}{0}$ $\frac{S(H)}{1}$



1000000
1010100

S	R	Q_{n+1}
0	0	Q_n Hold
0	1	0 Reset
1	0	1 Set
1	1	0 Reset

$Q_n \rightarrow Q_{n+1}$	S	R
0 0	0 0	0 0
0 1	0 1	0 0
1 0	0 0	1 0
1 1	0 0	0 0

THE NOR-CENTERED BASIC CELL IS
RESET DOMINANT

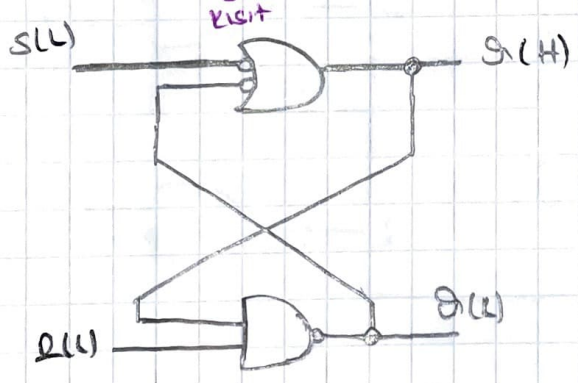
Basin olan 0 0 yaden

BASIC CELL : GENERIC FORM

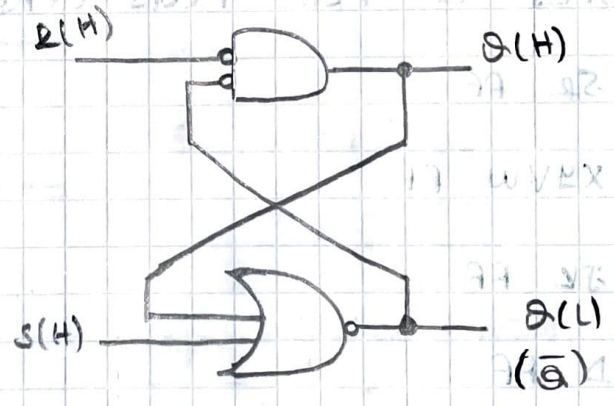
S	R	Q_{n+1}
0	0	Q_n (Hold)
0	1	0 (Reset)
1	0	1 (Set)
1	1	0

$Q_n \rightarrow Q_{n+1}$	S	R
0 0	0	0
0 1	1	0
1 0	0	1
1 1	0	0

constraint $\Rightarrow S.R = 0$

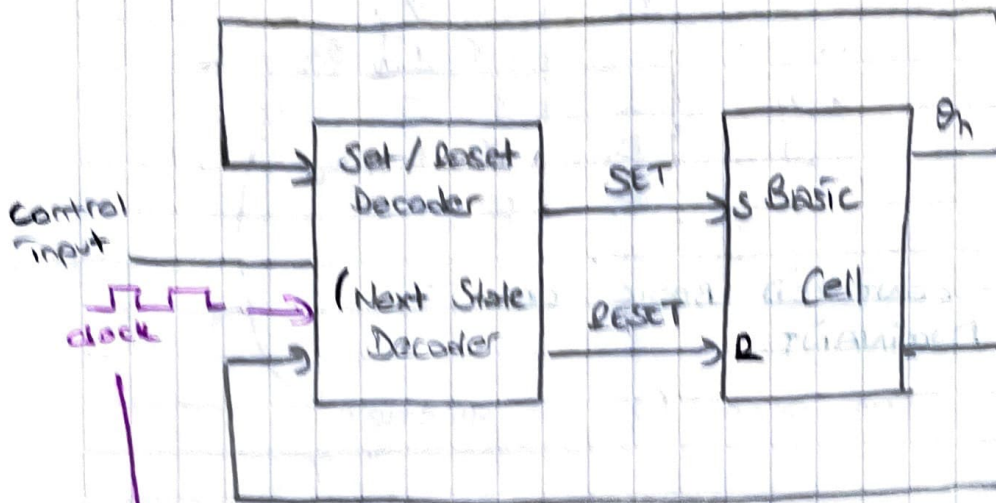


NAND BASIC CELL

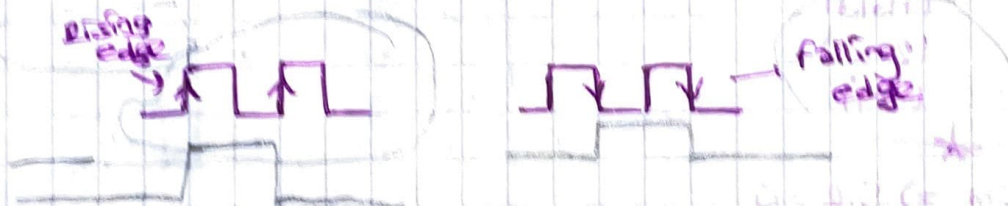


NOR BASIC CELL

FLIP - FLOPS



temel here flip-flop' to tetiklene kerana yok



TYPES OF FLIP - FLOPS (FFs)

- ① SR FF
- ② XYVW FF
- ③ JK FF
- ④ D FF
- ⑤ T FF
- ⑥ Master - Slave FF

① S-R FF



Mercurat durum $\uparrow$ serbest deyerler

özet/ **state**

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	$\emptyset$
1	1	1	$\emptyset$

Hold $\rightarrow$ Hold
 Reset $\rightarrow$ " "
 set $\rightarrow$ set

Don't care durumlar

$Q_n \rightarrow Q_{n+1}$	S	R
0 0	0	$\emptyset$
0 1	1	0
1 0	0	1
1 1	$\emptyset$	0

$$(Klob) Q_{n+1} = S + \bar{R} \cdot Q_n$$

Önceki satır 1 0'a geçirmek için tek bu.

$$S \cdot R = 0$$

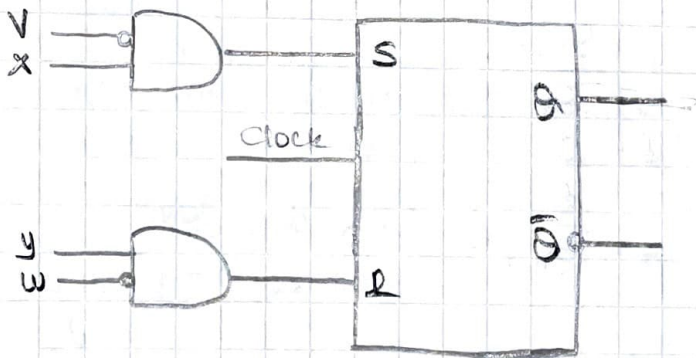
Durum Geçiş Tablosu

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
1	0	0	1
1	0	1	1

$\bar{R} \cdot Q_n$
 $S + \bar{R} \cdot Q_n$

② XYVW FF

Teorik Flip Flop



$$S = X \cdot \bar{V}$$

$$R = Y \cdot \bar{W}$$

$$Q_{n+1} = S + \bar{R} \cdot Q_n$$

$$= X \cdot \bar{V} + \bar{Y} \cdot \bar{W} \cdot Q_n$$

$$S \cdot R = 0 \text{ (constraint)}$$

$$X \cdot Y \cdot \bar{V} \cdot \bar{W} = 0$$

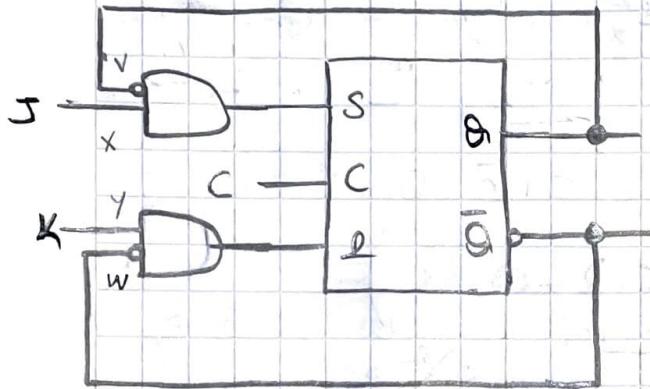
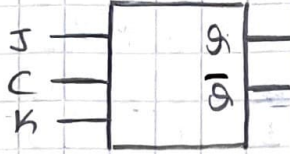
0 0 0 0
1 1 $\emptyset$ 0

→ Flip floplar C tipi bir makinedir

③ JK FF

$$X = T, \quad Y = K$$

$$V = Q, \quad W = \bar{Q}$$



set	reset		
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Hold

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

$$Q_{n+1} = Q_n$$

Q_n Hold

→ Mercurt durum kaynar
K ucu Resetleme görevi alır 0
J ucu set görevi alır 1
S-R den farkı

Sadece 1-1 durumu

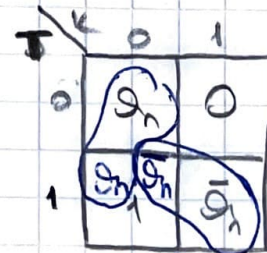
JK, 1-1'de
türünü
alıyor.

Geçiş
Tablosu

Transition
Table

$$Q_{n+1} = J \cdot \bar{Q}_n + \bar{K} \cdot Q_n$$

$$JK \cdot Q_n \bar{Q}_n = 0$$



$$\bar{K} \cdot Q_n + J \cdot \bar{Q}_n$$

→ Hocam bunu sevdi

Geciktirme sagliyor.
Temel hafiza elemanı

(Data FF) Data Flip Flop

④ D FF

→ Temel bellek elemanı olarak kullanılır
→ Geciktirme elemanı olarak

$$X = Y = 1$$

$$W = \bar{V} = D$$

$$Q_{n+1} = D + D \cdot Q_n = D$$

$$Q_{n+1} = D$$

önceki durum

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

D	Q_{n+1}
0	0
1	1



Q_n önemli değil
rise $Q_{n+1} = D$
D 1 ise $Q_{n+1} = 1$

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Birbirinin aynı $Q_{n+1} = D$

⑤ T FF

(Toggle Flip Flop)
(Toggle FF)

$$J = K = T$$

$$Q_{n+1} = T \cdot \bar{Q}_n + \bar{T} \cdot Q_n$$

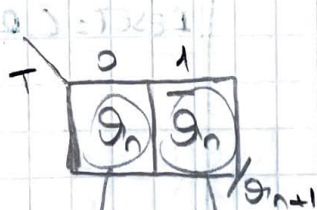
$$Q_{n+1} = T \oplus Q_n$$

No constraint

T	Q_n	Q_{n+1}
0	0	0 (hold)
0	1	1
1	0	1
1	1	0

T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



$$Q_n \cdot \bar{T} + \bar{Q}_n \cdot T$$

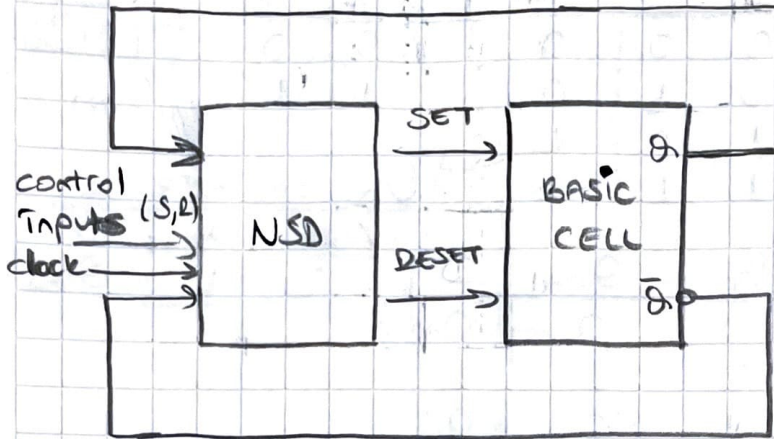
$$Q_{n+1} = T \oplus Q_n$$

0 durumunda
TFF pozitif
Q neyse Q_{n+1} 'a
aykırı

→ Hoca senyor!!

THE DESIGN OF FF'S

→ S L Flip Flop tasarlayalım



uygum tablosu

S	L	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Ø

NSD'ye koy

Q_n	Q_{n+1}	S	L
0	0	0	Ø
0	1	1	0
1	0	0	1
1	1	Ø	0

NSD'ye koy

NSD (Next State Decoder'in Doğruluk Tablosu)

* clk "0" 1
0'dur ya da 2
Q = Q_{n+1}

	CLK	S	L	Q_n	Q_{n+1}	SET	RESET
0	0	0	0	0	0	0	Ø
1	0	0	0	1	1	Ø	Ø
2	0	0	1	0	0	0	Ø
3	0	0	1	1	1	Ø	Ø
4	0	1	0	0	0	0	Ø
5	0	1	0	1	1	Ø	Ø
6	0	1	1	0	0	0	Ø
7	0	1	1	1	1	Ø	Ø
8	1	0	0	0	0	0	Ø
9	1	0	0	1	1	Ø	Ø
10	1	0	1	0	0	0	Ø
11	1	0	1	1	0	0	1
12	1	1	0	0	1	1	0
13	1	1	0	1	1	Ø	Ø
14	1	1	1	0	Ø	Ø	Ø
15	1	1	1	1	Ø	Ø	Ø

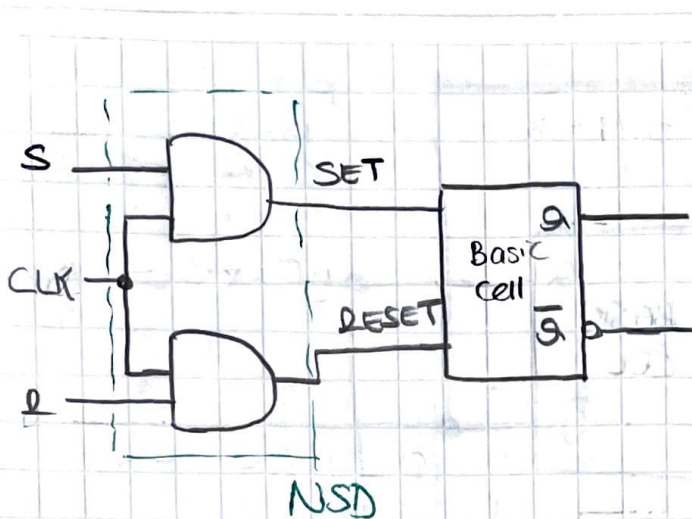
⇒ k.map

SET = C.S
RESET = C.L

FF - Flip Flop'un doğruluk tablosu

S.R = 0 olmalı,

→ Gözetim tablosu çizilip



T FF

CS	Q _n	00	01	11	10
00	0	0	0	0	0
01	0	0	0	0	0
11	1	0	0	0	0
10	0	0	0	0	0

SET

$$SET = C.S$$

CS	Q _n	00	01	11	10
00	0	0	0	0	0
01	0	0	0	0	0
11	0	0	0	0	0
10	0	0	1	0	0

RESET

$$RESET = C.R$$

	CLK	T	Q _n	Q _{n+1}	SET	RESET
0	0	0	0	0	0	0
1	0	0	1	1	0	0
2	0	1	0	0	0	0
3	0	1	1	1	0	0
4	1	0	0	0	0	0
5	1	0	1	1	0	0
6	1	1	0	1	1	0
7	1	1	1	0	0	1

⇓

K-map

$$SET = C.T.\bar{Q}_n$$

$$RESET = C.T.Q_n$$

C	T	Q _n	00	01	11	10
0	0	0	0	0	0	0
1	0	0	0	0	1	0

SET

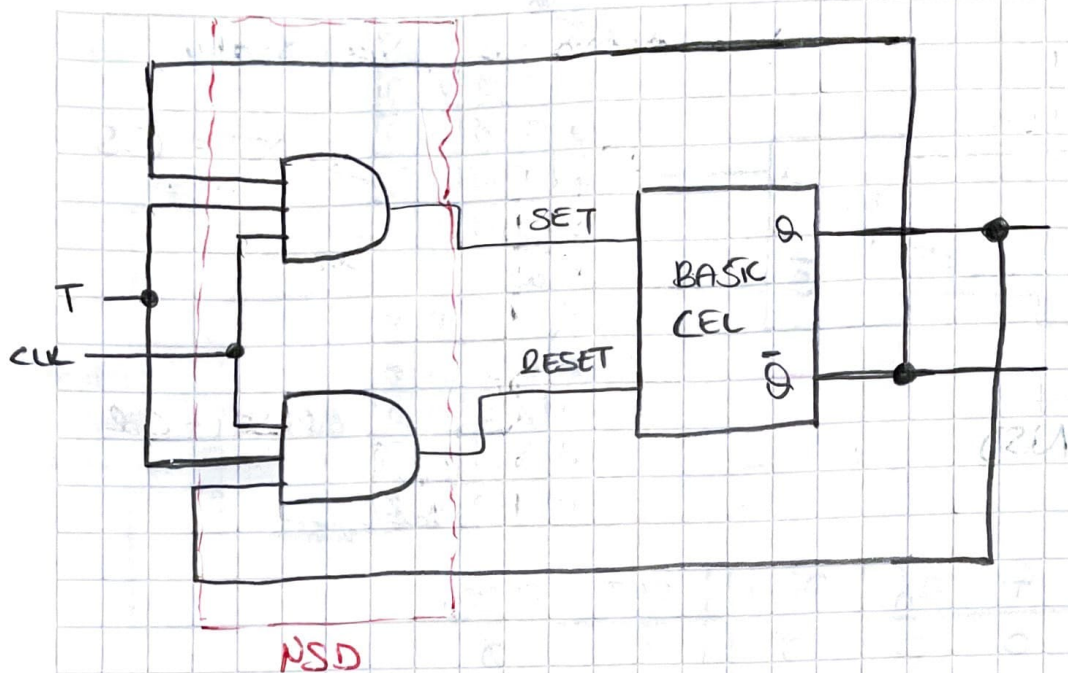
$$SET = C.T.\bar{Q}_n$$

C	T	Q _n	00	01	11	10
0	0	0	0	0	0	0
1	0	0	0	1	0	0

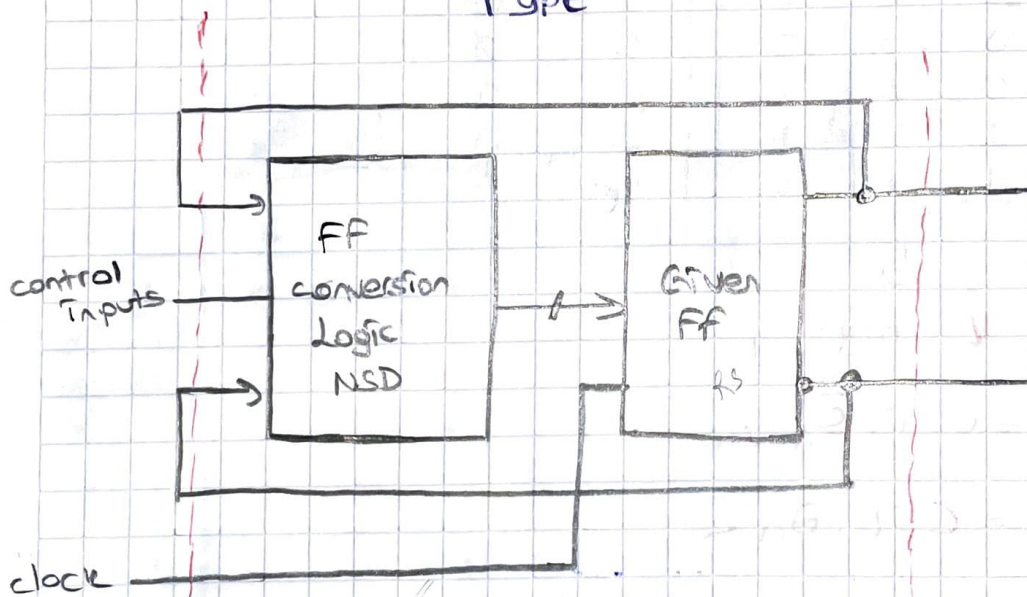
RESET

$$C.T.Q_n = RESET$$

→ sayıa konusu kritiki



FLIP-FLOP CONVERSION FROM ONE TYPE TO ANOTHER TYPE



Desired FF

JK

J K

J K Q

Qn-1

elde etmek istediğimiz Q_{n+1} bazaya göre

SR Flip-Flop'undan D Flip-Flop'a elde edelim

a) $SR \rightarrow D$

b) $D \rightarrow T$

$$\begin{array}{c|cc} D/Q_n & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 1 & \emptyset \end{array}$$

$S = D$

$$\begin{array}{c|cc} D/Q_n & 0 & 1 \\ \hline 0 & \emptyset & 1 \\ 1 & 0 & 0 \end{array}$$

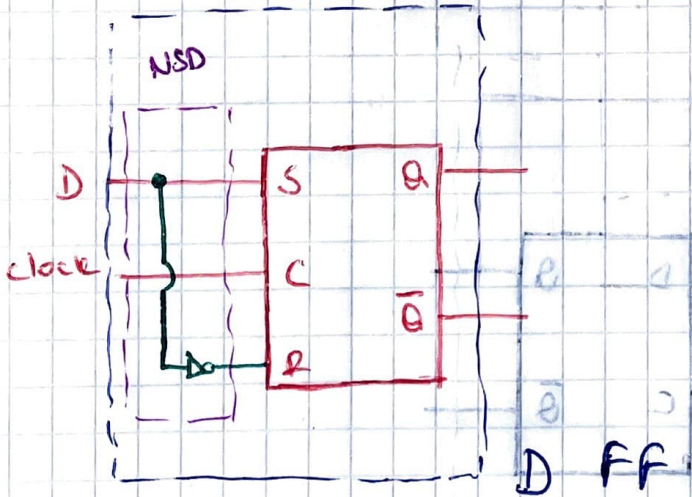
$R = \bar{D}$

(2)

D	Q_n	Q_{n+1}	S	R
0	0	0	0	0
1	0	1	0	1
2	1	0	1	0
3	1	1	0	0

K-map

$S = D$
 $R = \bar{D}$



D	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

⑥

T	a_n	a_{n+1}	D
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

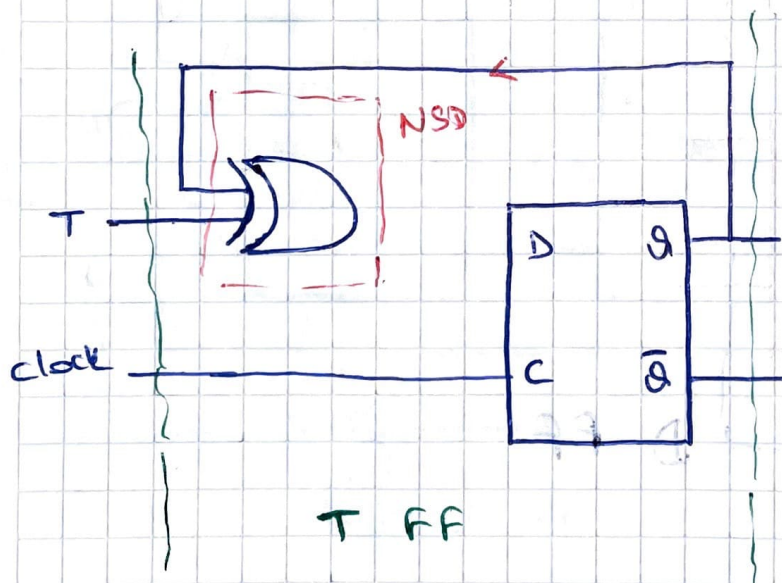
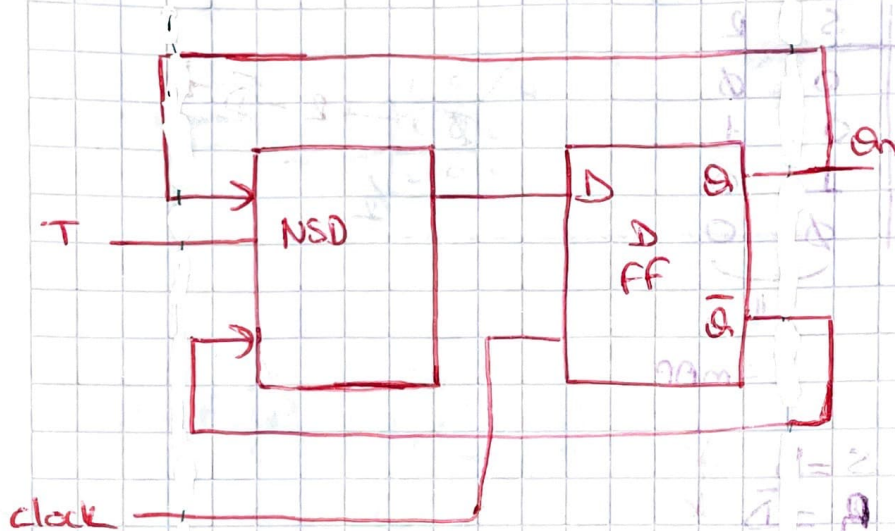
k-map
 $D = T \oplus a_n$

D	a_{n+1}
0	0
1	1

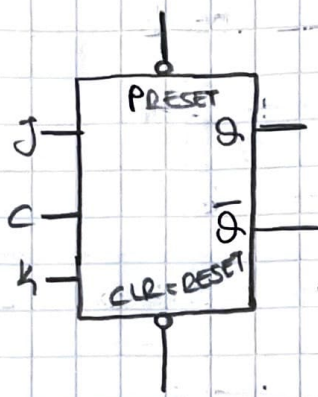
T	a_n	D
0	0	0
0	1	1
1	0	1
1	1	0

$D = \bar{T} \cdot a_n + T \cdot \bar{a}_n$

$D = T \oplus a_n$

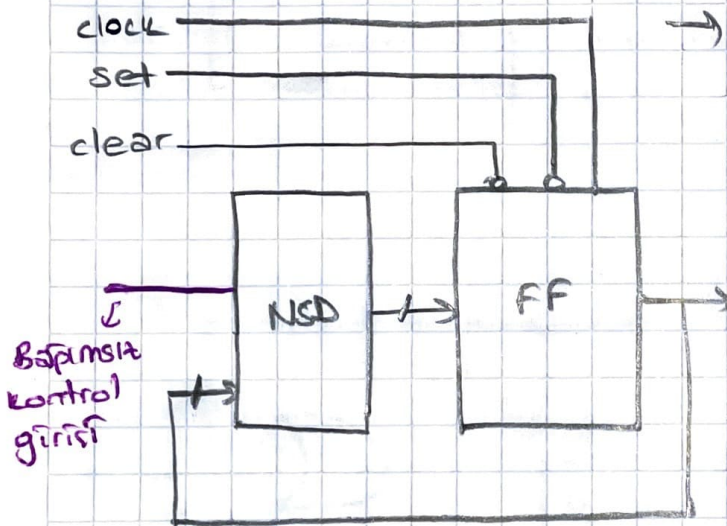


FF with clear and preset inputs



(preset goes veriyor)
+5V

COUNTERS



→ Sayıcılar değere olmuş C tipi bir makinedir.

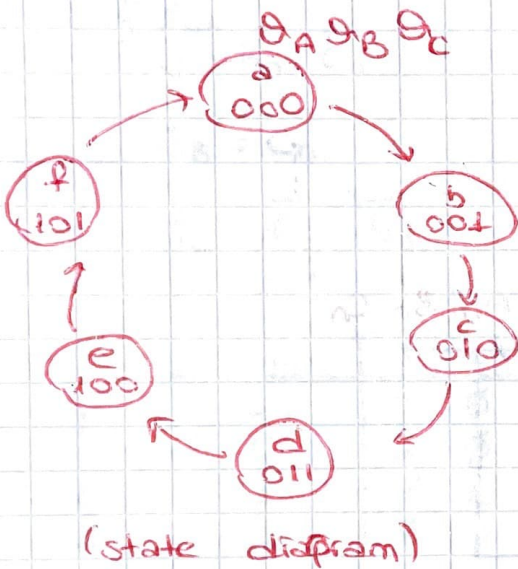
multimode machine

5 Criteria

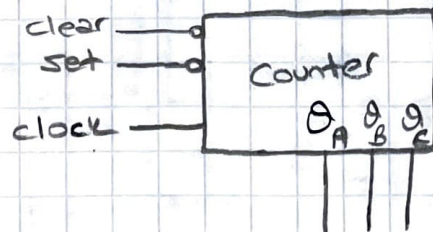
- 1) Single mode, multimode
- 2) Number of output bits
- 3) Module number → kararlı durum → mod 7, mod 10
sayısı
↓ ↓
0'dan 6'ya 0'dan 9'a
kadar sayar kadar sayar
- 4) Code Sequence
- 5) Synchronous - Asynchronous

Single mode Counter Design

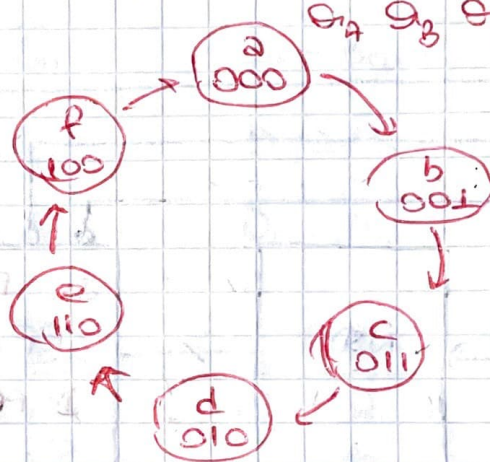
- Single mode
- Three bits
- mod 6
- Binary code counter
- Synchronous



Binary Code



- Single mode
- Three bits
- mod 6
- Unit distance code
- Synchronous



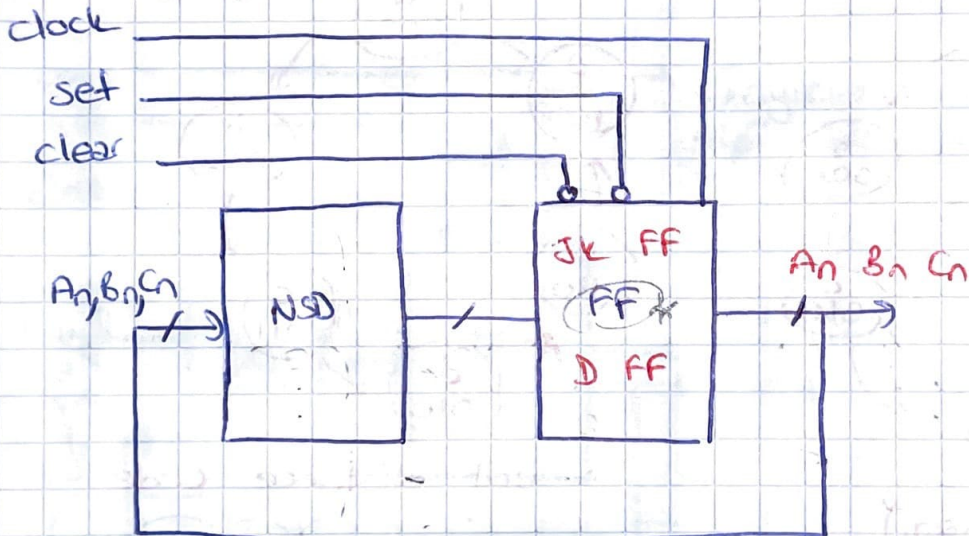
unit distance code

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10

204 bit rose o
leader flip flop

dec	A_n			A_{n+1}		
	A_n	B_n	C_n	A_{n+1}	B_{n+1}	C_{n+1}
0	0	0	0	0	0	1
1	0	0	1	0	1	1
3	0	1	1	0	1	0
2	0	1	0	1	1	0
6	1	1	0	1	0	0
4	1	0	0	0	0	0
5	1	0	1	$\emptyset$	$\emptyset$	$\emptyset$
7	1	1	1	$\emptyset$	$\emptyset$	$\emptyset$

kararlı durum değil
(don't care durumları)



JK FF
D FF

Dec	A _n	B _n	C _n	A _{n+1}	B _{n+1}	C _{n+1}	S J _A	R K _A	J _B	K _B	J _C	K _C	D _A	D _B	D _C
0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	1
1	0	0	1	0	1	1	0	0	1	0	0	0	0	1	1
3	0	1	1	0	1	0	0	0	0	0	0	1	0	1	0
2	0	1	0	1	1	0	1	0	0	0	0	0	1	1	0
6	1	1	0	1	0	0	0	0	0	1	0	0	1	0	0
4	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
5	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
7	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0

A _n	B _n	C _n	011	110
0	0	0	0	1
1	0	0	0	0

A _n	B _n	C _n	011	110
0	0	0	0	0
0	1	0	0	0

⏚ K - map

$J_A = B_n \cdot \bar{C}_n$

$K_A = \bar{B}_n$

$D_A = B_n \cdot \bar{C}_n$

$J_B = C_n$

$K_B = A_n$

$D_B = \bar{A}_n \cdot B_n + C_n$

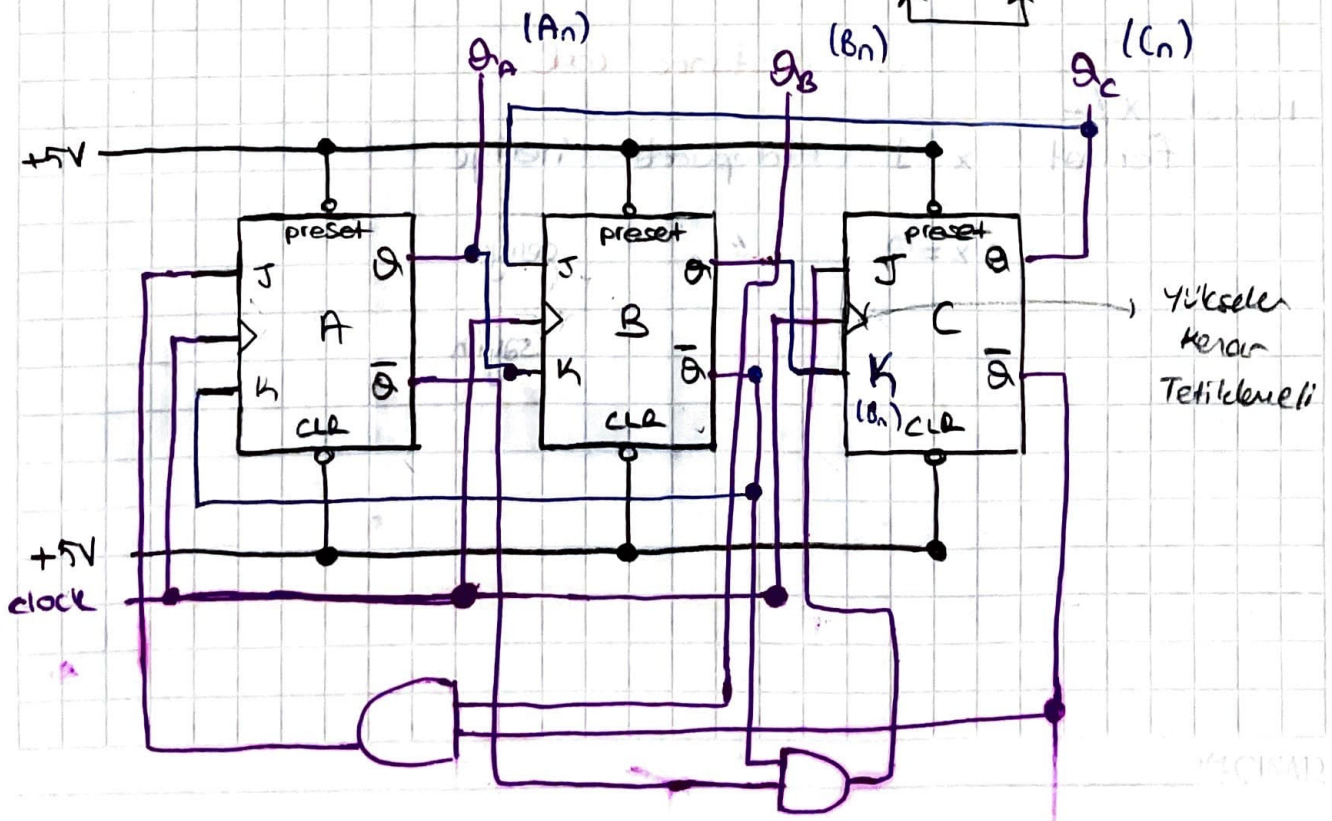
$J_C = \bar{A}_n \cdot \bar{B}_n$

$K_C = B_n$

$D_C = \bar{A}_n \cdot \bar{B}_n$

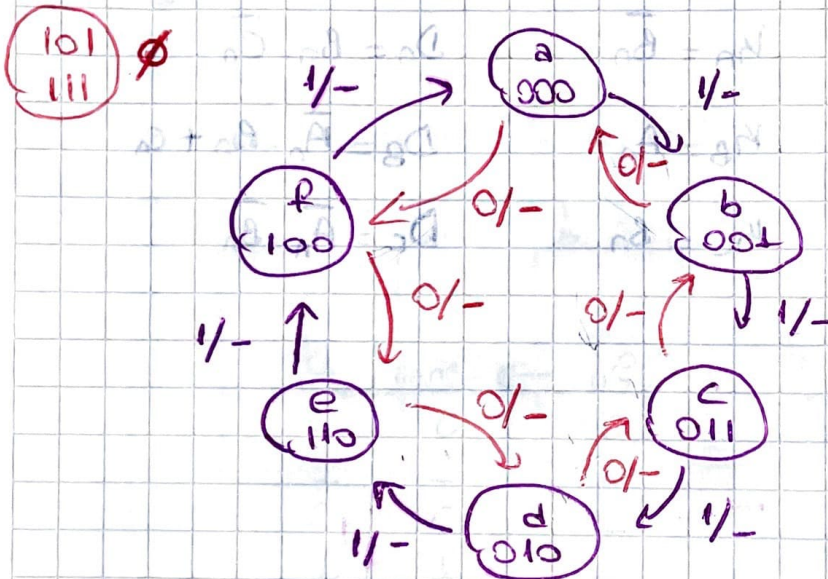
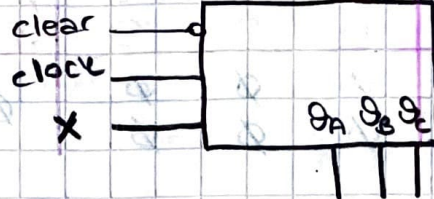
A _n → A _{n+1}	J	K
0 → 0	0	0
0 → 1	1	0
1 → 0	0	1
1 → 1	0	0

A _n → A _{n+1}	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1



MULTIMODE COUNTER

- Three bits
- Unit Distance Code
- Mod 6
- Synch.
- Up/Down

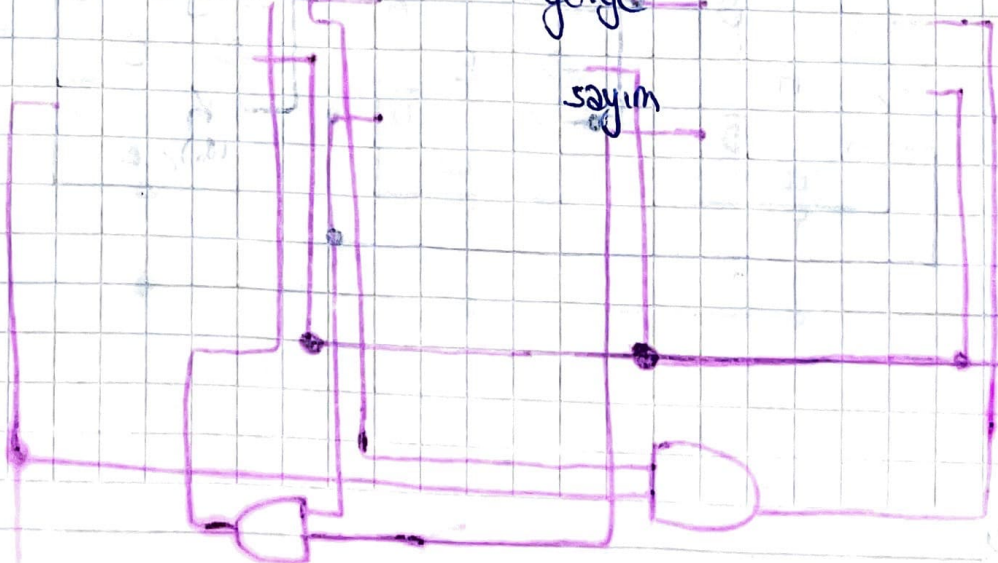


format x/-

Unit distance code

x=1 olduğunda ileriye
x=0 " geriye

sayım



D çıkışları S_{n+1} ile aynı !!!

Binary
Değerleri bozma
 D_n S_{n+1} D_c

DA DB DC bunların aynı Transition table'i yi boyay

	A_n	B_n	C_n	X	A_{n+1}	B_{n+1}	C_{n+1}	J_A	J_B	J_C	K_B	J_C	K_C
0	0	0	0	0	1	0	0	1	0	0	0	0	0
1	0	0	0	1	0	0	1	0	0	0	0	1	0
2	0	0	1	0	0	0	0	0	0	0	0	0	1
3	0	0	1	1	0	1	1	0	0	1	0	0	0
4	0	1	0	0	0	1	1	0	0	0	0	1	0
5	0	1	0	1	1	1	0	1	0	0	0	0	0
6	0	1	1	0	0	0	1	0	0	0	1	0	0
7	0	1	1	1	0	1	0	0	0	0	0	0	1
8	1	0	0	0	1	1	0	0	0	1	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0	0	0
10	1	0	1	0	0	0	0	0	0	0	0	0	0
11	1	0	1	1	0	0	0	0	0	0	0	0	0
12	1	1	0	0	0	1	0	0	1	0	0	0	0
13	1	1	0	1	1	0	0	0	0	0	1	0	0
14	1	1	1	0	0	0	0	0	0	0	0	0	0
15	1	1	1	1	0	0	0	0	0	0	0	0	0

X'i en solda atarsam

data tablosu (4x16 decoder)

$$D_A = \bar{C}_n \cdot (B_n \odot X)$$

$$D_B = A_n \cdot \bar{X} + \bar{A}_n \cdot B_n \cdot \bar{C}_n + C_n \cdot X$$

$$D_C = \bar{A}_n \cdot (B_n \oplus X)$$

$$J_A = \bar{C}_n \cdot (B_n \odot X)$$

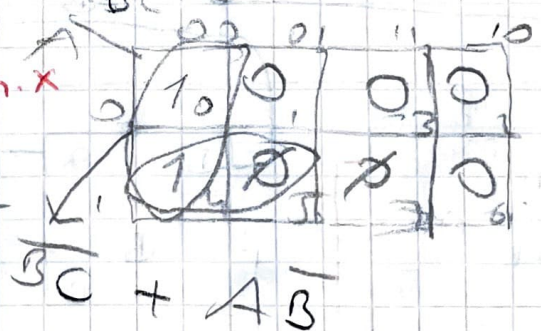
$$K_A = B_n \oplus X$$

$$J_B = A_n \cdot \bar{X} + C_n \cdot X$$

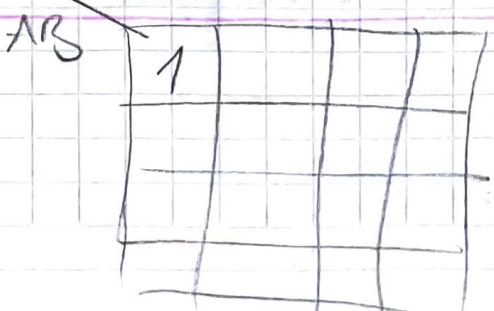
$$K_B = A_n \cdot \bar{X} + C_n \cdot X$$

$$J_C = \bar{A}_n \cdot (B_n \oplus X)$$

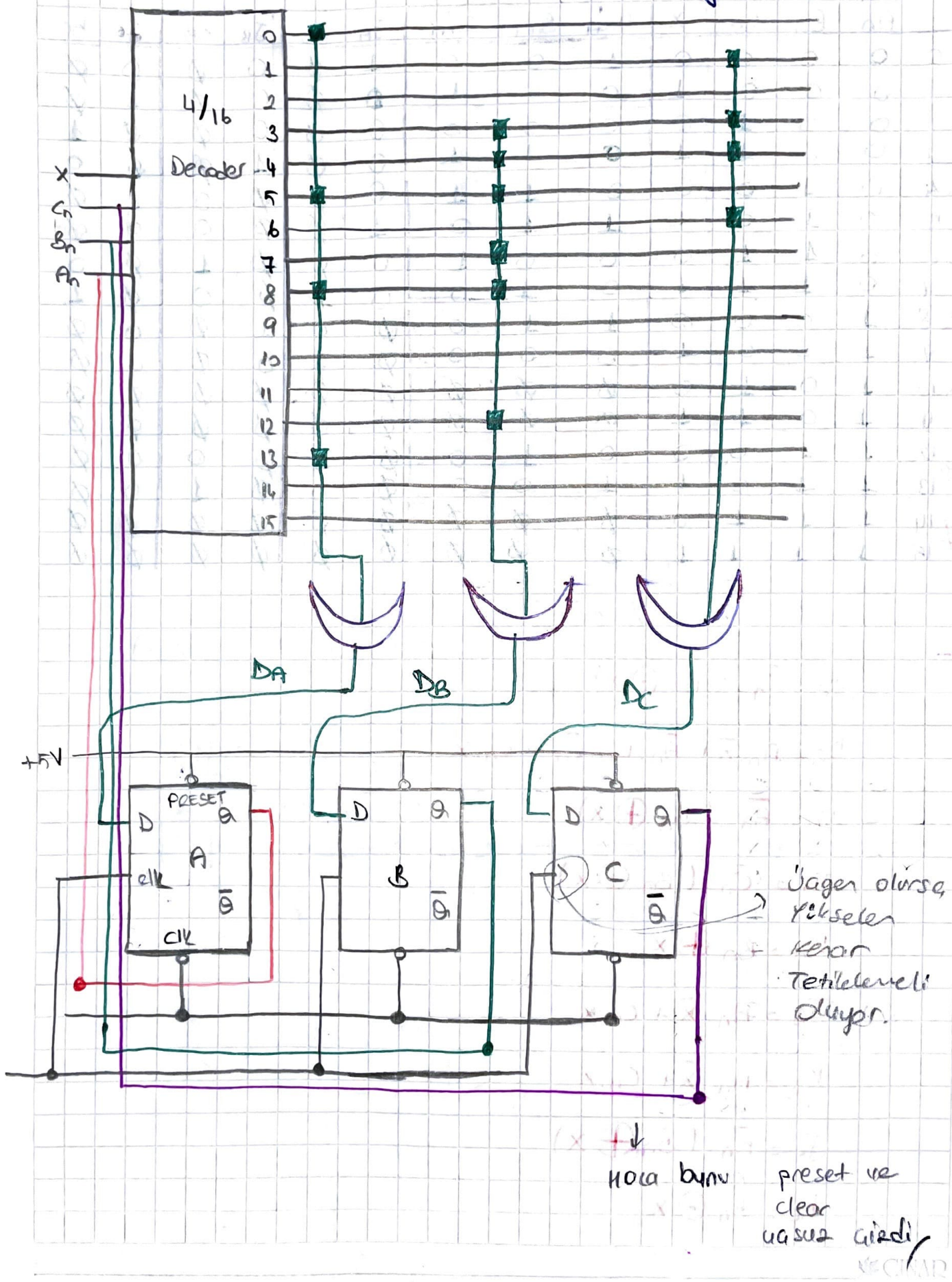
$$K_C = B_n \odot X$$



$$\bar{B} (A + \bar{C})$$



(4x16 Decoder nie
gezeichnet)



RING COUNTERS (JOHNSON OR MÖBIUS COUNTER)

① Standard - Ring Code → Counter

② Twisted - Ring Code → Counter

① Standard Ring Code (One Hot Counters)

MSB					LSB				
	A	B	C	D		A	B	C	D
a	0	0	0	0		0	0	0	0
b	1	0	0	0		0	0	0	1
c	0	1	0	0		0	0	1	0
d	0	0	1	0		0	1	0	0
e	0	0	0	1		1	0	0	0
	-	-	-	-		-	-	-	-
tekrar eden kısım	0	0	0	0		0	0	0	0

Solara msb veya lsb (bit) evrilip kaydırılıyor.

② Twisted Ring Code (Johnson Counter, Möbius Counter)

MSB				LSB			
A	B	C		A	B	C	
0	0	0		0	0	0	
1	0	0		0	0	1	
1	1	0		0	1	1	
1	1	1		1	1	1	
0	1	1		1	1	0	
0	0	1		1	0	0	
-	-	-		-	-	-	
tekrar eden kısım	0	0	0	0	0	0	

6 kararlı durum var

msb evirmeli

lsb evirmeli

Bunda ise lsb veya msb'den başlayarak her seferinde bir sonraki bit evriliyor. Hepsini 1 olduğunda tekrar aynı evrilme işlemi yapılıyor.

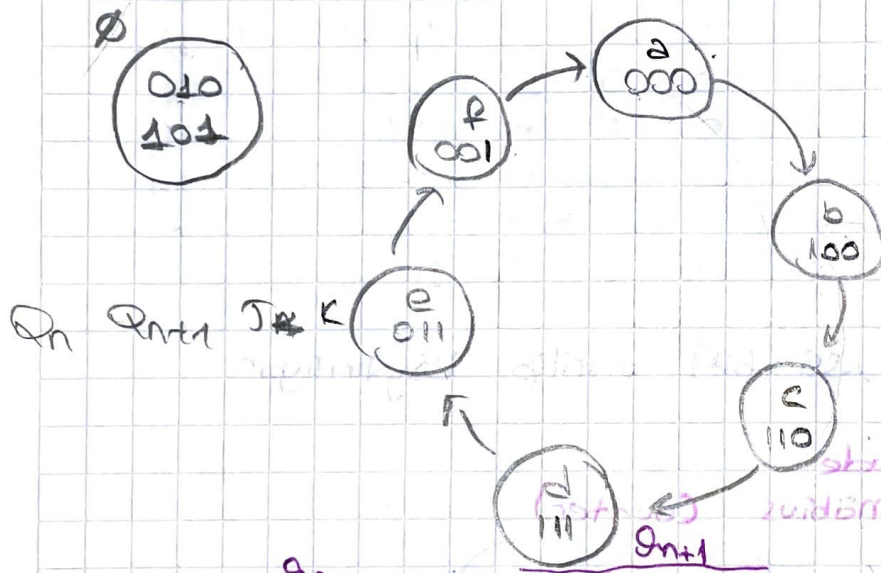
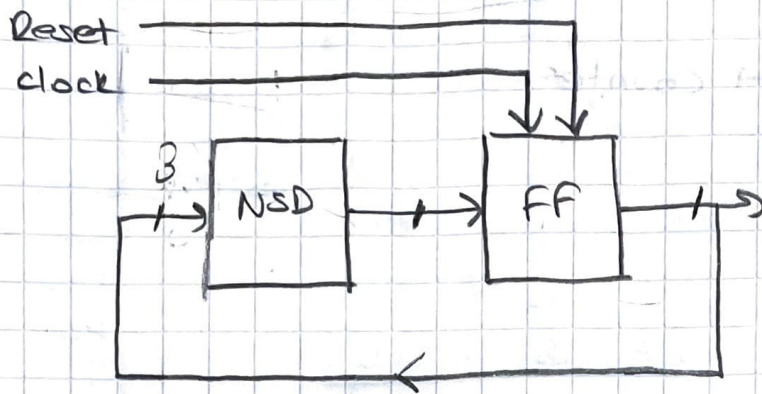
Ö21 3 bit twisted ring counter tasarlayalım.
(msb evirmeli olsun)

evirmeli:

sop

0'lar

max



A	B	C	A	B	C
0	0	0	0	0	0
1	0	1	0	1	0
2	0	0	1	0	0
3	1	0	1	1	0
4	1	1	0	1	1
5	0	1	1	0	1
6	0	1	0	0	1
7	0	1	1	0	0

	A_n	B_n	C_n	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	1	0	0	0	0	0
1	0	0	1	0	0	0	0	0	1
2	0	1	0	0	0	0	0	0	0
3	0	1	1	0	0	0	1	0	0
4	1	0	0	1	1	0	1	0	0
5	1	0	1	0	0	0	0	0	0
6	1	1	0	1	1	0	0	1	0
7	1	1	1	0	1	0	0	0	0

NSD for DFF

NSD for JK FF

